Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A solution containing 2.675 g of $CoCl_3.6NH_3$ (molar mass $= 267.5$ g mol^{-1}) is passed through a cation exchanger. the chloride ions obtained is solution were treated with excess of $AgNO_3$ to give 4.78 g of $AgCl_{(\text{molar mass } = 143.5 \text{ g mol}}$ (At. mass). The formula of the complex is

(A) $[CoCl_3(NH_3)_3]$

(B) $[CoCl(NH_3)_5]Cl_2$

(C) $[Co(NH_3)_6]Cl_3$

(D) $[CoCl_2(NH_3)_4]Cl$

Q.2

Rusting of iron in moist air involves-

(A) Loss of electrons by iron

(B) Gain of electrons by iron

(C) Neither gain nor loss of electrons

(D) Hydration of iron

Q.3

When copper oxide is treated with sulphuric acid it produces a compound [X] which exist as blue coloured aqueous solution when it is heated its colour changes to white. When [X] is treated with excess of aqueous NH_3 it again produces a deep blue coloured solution of [Y].
Identify the correct statement about [X] and [Y].

(A) [X] is $Cu(SO_4)\cdot 5H_2O$ and [Y] is $[Cu(NH_3)_6]^{2+}$

(B) [X] is $[Cu(H_2O)_4]SO_4\cdot H_2O$ and [Y] is $[Cu(NH_3)_4]SO_4$

(C) [X] is $[Cu(H_2O)_4]SO_4\cdot H_2O$ and [Y] is $[Cu(NH_3)_4(H_2O)_2]^{2+}$

(D) [X] is $[Cu(H_2O)_6]SO_4$ and [Y] is $[Cu(NH_3)_4]SO_4$

Q.4

Zn and Hg do not show variable valency like d-block elements because-

(A) They are soft

(B) Their d-shells are complete

(C) They have only two electrons in the outermost subshell

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(D) Their d-shells are incomplete

Q.5

Which of the following statements is incorrect?

(A) The compoounds formed by the corresponding $3\ \mathrm{d}$ transition metals in lower valence states are ionic but those in higher valence state are covalent

(B) The $4\ \mathrm{d}$ and $5\ \mathrm{d}$ transition metals form less ionic compounds than the $3\ \mathrm{d}$ transiton metals

(C) The compounds formed by the $3d$ transiton metals are less ionic than the corresponding compounds formed by the $4\ \mathrm{d}$ and $5\ \mathrm{d}$ transition metals

(D) The ionization energies of $3\ \mathrm{d}, 4\ \mathrm{d}$ and $5\ \mathrm{d}$ transiton metals are greater than those of group 1 and 2 metals

Q.6

Out of the compounds $\mathrm{K_2SO_4}, \mathrm{MgCl_2}, \mathrm{FeSO_4}, \mathrm{NiCl_2}$ and ZnO which of the following pair will show paramagnetism-

(A) $\mathrm{K_2SO_4}, \mathrm{MgCl_2}$

(B) $\mathrm{ZnO}, \mathrm{MgCl_2}$

(C) $\mathrm{K_2SO_4}, \mathrm{ZnCl_2}$

(D) $\mathrm{FeSO_4}, \mathrm{NiCl_2}$

Q.7

Amount of oxalic acid present in a solution can be determined by its titration with $\mathrm{KMnO_4}$ solution in the presence of $\mathrm{H_2SO_4}$. The titration gives unsatisfactory result when carried out in the presence of HCl, because HCl

(A) gets oxidised by oxalic acid to chlorine

(B) furnished $\mathrm{H^+}$ ions in addition to those from oxalic acid

(C) reduces permanganate to $\mathrm{Mn^{2+}}$

(D) Oxidises oxalic acid to carbon dioxide and water

Q.8

An ion $\mathrm{M^{2+}}$, forms the complexes $\left[\mathrm{M\left(H_2O\right)_6}\right]^{2+}$, $\left[\mathrm{M(en)_3}\right]^{2+}$ and $\left[\mathrm{MBr_6}\right]^{4-}$, match the complex with the appropriate colour.

(A) Green, blue and red

(B) Blue, red and green

(C) Green, red and blue

(D) Red, blue and green

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Q.9

$0.001\ \text{mol}$ of $\text{Co}\left(\text{NH}_3\right)_5\left(\text{NO}_3\right)\left(\text{SO}_4\right)$ was passed through a cation exchanger and the acid coming out of it required 20ml of 0.1MNaOH for neutralisation. Hence the complex is :

(A) $\left[\text{CoSO}_4\left(\text{NH}_3\right)_5\right]\text{NO}_3$

(B) $\left[\text{CoNO}_3\left(\text{NH}_3\right)_5\right]\text{SO}_4$

(C) $\left[\text{Co}\left(\text{NH}_2\right)_2\right]\text{SO}_3\text{NO}_2$

(D) None of these

Q.10

All the following complexes show decrease in their weights when placed in a magnetic balance then the group of complexes having tetrahedral geometry is :

(i) Ni(CO)_4

(ii) $\text{K}\left[\text{AgF}_4\right]$

(iii) $\text{Na}_2\left[\text{Zn(CN)}_4\right]$

(iv) $\text{K}_2\left[\text{PtCl}_4\right]$

(v) $\left[\text{RhCl}_{(\text{PPh}_3)_3}\right]$

(A) (ii), (iii), (v)

(B) (i), (ii), (iii)

(C) (i), (iii), (iv)

(D) None of these

Q.11

Among the following ions which one has the highest paramagnetism

(A) $\left[\text{Cr}\left(\text{H}_2\text{O}\right)_6\right]^{3+}$

(B) $\left[\text{Fe}\left(\text{H}_2\text{O}\right)_6\right]^{2+}$

(C) $\left[\text{Cu}\left(\text{H}_2\text{O}\right)_6\right]^{2+}$

(D) $\left[\text{Zn}\left(\text{H}_2\text{O}\right)_6\right]^{2+}$

Q.12

The compound which does not show paramagnetism is

(A) $\left[\text{Cu}\left(\text{NH}_3\right)_4\right]\text{Cl}_2$

(B) $\left[\text{Ag}\left(\text{NH}_3\right)_2\right]\text{Cl}$

(C) NO

(D) NO_2

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Q.13

What will be the 'spin only' magnetic moment of the complex formed when $\mathrm{Fe(SCN)_3}$ reacts with a solution containing excess $\mathrm{F^-}$?

(A) 2.83 B.M.

(B) 3.87 B.M.

(C) 5.92 B.M.

(D) 1.73 B.M.

Q.14

Which one of the following statements concerning lanthanides elements is/are true -

(A) Lanthanides are separated from one another by ion exchange methods

(B) The ionic radii of trivalent lanthanides steadily increase with increase in atomic number

(C) All lanthanides are highly dense metals

(D) Most typical oxidation of lanthanides is $+3$

Q.15

Cerium $(Z = 58)$ is an important member of the lanthanoids. Which of the following statements about cerium is/are correct

(A) The $+4$ oxidation state of cerium is not known in solutions

(B) The $+3$ oxidation state of cerium is more stable than the $+4$ oxidation state

(C) The common oxidation states of cerium are $+3$ and $+4$

(D) Cerium (IV) acts as an oxidizing agent

Q.16

The compound $\mathrm{Na_2IrCl_6}$ reacts with triphenylphosphine in diethyleneglycol in an atmposhere of CO to give $\mathrm{[IrCl(CO)\,(Ph_3)_2]}$, thrown as 'Vaska's compound' (Atomic number of $\mathrm{Ir} = 77$)
Which of the following statements is/are correct -

(A) The IUPAC name of the complex is carbonlychloridobis (triphenylphosphine) iridium (I).

(B) The hybridisation of the metal ion is $\mathrm{sp^3}$.

(C) The magnetic moment (spin only) of the complex is zero

(D) The complex shows geometrical as well as ionization isomerism.

Q.17

Passage
Crystal field theory views the bonding in complexes as arising from electrostatic interaction and considers the effect

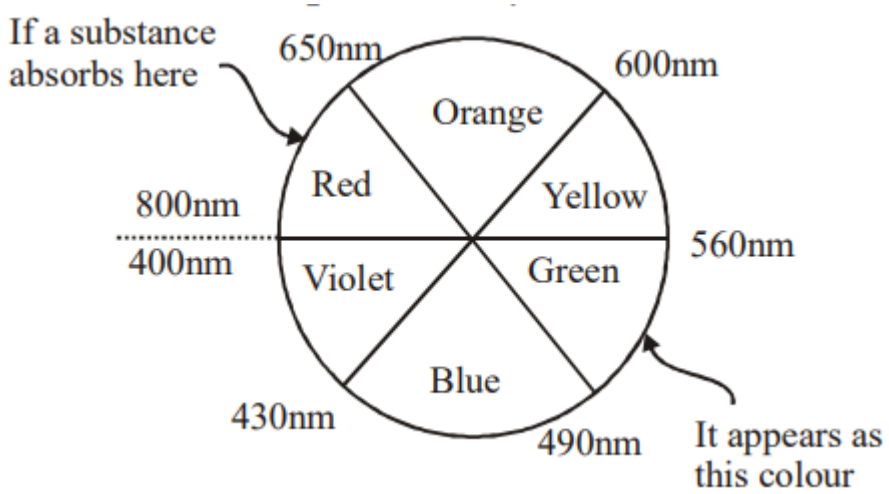
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of the ligand charges on the energies of the metal ion d-orbitals.

In this theory, a ligand lone pair is modelled as a point negative charge that repels electrons in the d-orbitals of the central metal ion. The theory concentrated on the resulting splitting of the dorbitals into two groups with different energies and used that splitting to rationalize and correlate the optical spectra, thermodynamic stability, and magnetic properties of complexes. This energy splitting between the two sets of d-orbitals is called the crystal field splitting Δ .

In general, the crystal field splitting energy Δ corresponds to wavelength of light in visible region of the spectrum, and colours of the complexes can therefore be attributed to electronic transition between the lower and higher energy sets of d-orbitals.

In general, the colour that we see is complementary to the colour absorbed.



Different metal ion have different values of Δ, which explains why their complexes with the same ligand have different colour.

Similarly the crystal field splitting also depends on the nature of ligands and as the ligand for the same metal varies from H_2O to NH_3 to ethylenediamine, Δ for complexes increases. Accordingly, the electronic transition shifts to higher energy (shorter wavelength) as the ligand varies from H_2O to NH_3 to en, thus accounting for the variation in colour.

Crystal field theory accounts for the magnetic properties of complexes in terms of the relative values of Δ and the spin pairing energy P. Small Δ values favour high spin complexes, and large Δ values favour low spin complexes.

Question

The $[Ti(NCS)_6]^{3-}$ ion exhibits a single absorption band at $544\ nm$. What will be the crystal field splitting energy (in $kJmol^{-1}$) of the complexes ?

$(h = 6.626 \times 10^{-34} Js, C = 3.0 \times 10^8\ m/s, N_A = 6.02 \times 10^{23}$ ions $/mole)$

(A) 240

(B) 220

(C) 270

(D) 250

Q.18

Passage

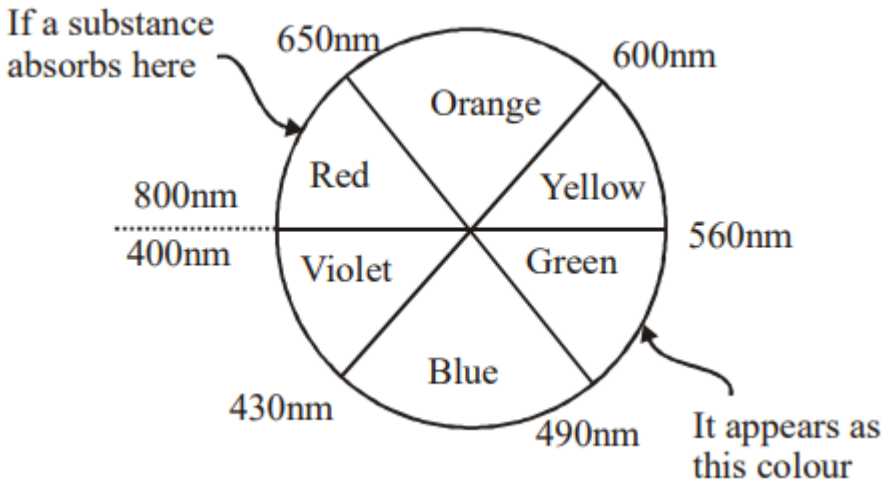
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Question

Which of the following statements is incorrect -

(A) The Ni^{2+} (aq) cation is coloured because Ni^{2+} ion can absorb light, which promotes electrons from the filled d-orbitals to the higher energy half filled d-orbitals.

(B) The Zn^{2+} (aq) cation is colourless because the d-orbitals are completely filled and no electrons can be promoted, so no light is absorbed.

(C) A complex which has just one absorption band at $455\ nm$, must be red coloured

(D) None of these

Q.19

Passage

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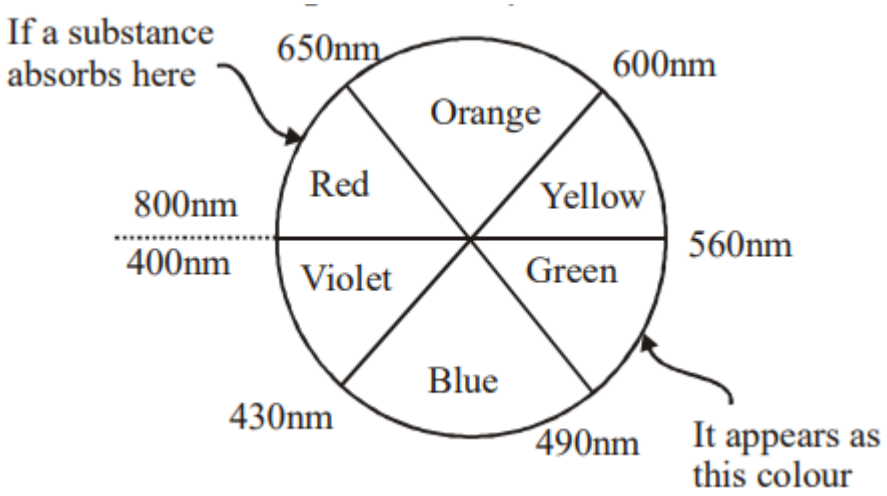
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Question

Which of the following complexes are diamagnetic

$[Pt(NH_3)_4]^{2+}$	$[Co(SCN)_4]^{2-}$	$[Cu(en)_2]^{2+}$	$[HgI_4]^{2-}$
Square planar	tetrahedral	square planar	tetrahedral
(i)	(ii)	(iii)	(iv)

(A) (i) \& (ii)

(B) (ii) \& (iii)

(C) (i) \& (iv)

(D) (iii) \& (iv)

Q.20

Passage

$Co^{2+}(aq) + SCN^-(aq) \rightarrow$ Complex (X)

$Ni^{2+}(aq)+$ Dimethylglyoxide $\xrightarrow{NH_4OH}$ Complex (Y)

The coordination number of cobalt and nickel in complexes X and Y are four.

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Question

The IUPAC names of the complexes (X) and (Y) are respectively -

(A) tetrathiocyanato-S-cobalt (II) and bis (dimethylglyoximate) nickel (II)

(B) tetrathiocyanato-S-cobaltate (II) and bis (dimethylglyoximato) nickel (II)

(C) tetrathiocyanato-S-cobaltate (II) and bis (dimethylglyoximato) nickelate (II)

(D) tetrathiocyanato-S-cobaltate (III) and bis (dimethylglyoximato) nickel (II)

Answers & Solutions

Q.1 Answer:

$[\mathrm{Co}\left(\mathrm{NH}_3\right)_6]\,\mathrm{Cl}_3$

Solution:

Moles of the complex $= \frac{2.675}{267.5} = 0.01$
moles of AgCl precipitated $= \frac{4.78}{143.5} = 0.033$
Thus, 1 mole of the complex will precipitate $\mathrm{AgCl}\frac{0.033}{0.01} = 3$ **moles.**
That means 1 molecule of the complex contains 3 ionizable Cl**.**
Hence, the formula is $[\mathrm{Co}\left(\mathrm{NH}_3\right)_6]\,\mathrm{Cl}_3$

Q.2 Answer:

Loss of electrons by iron

Solution:

Rusting of iron in moist air involves the loss of electrons by iron. The oxidation of iron to ferrous ion takes place and a brown deposit takes place on the surface of iron.
$\mathrm{Fe} \rightarrow \mathrm{Fe}^{+2} + 2\mathrm{e}$
Rusting is an electrochemical process.

Q.3 Answer:

[X] is $[\mathrm{Cu}(\mathrm{H_2O})_4]\mathrm{SO_4} \cdot \mathrm{H_2O}$ and [Y] is $[\mathrm{Cu}(\mathrm{NH_3})_4]\mathrm{SO_4}$

Solution:

This problem includes conceptual mixing of preparation and properties of copper sulphate.

To solve this problem student must go through reading the question and making the correct picture about characteristics of the molecule.

Preparation of Copper Sulphate $Cu(SO)_4$

When copper sulphate is treated with sulphuric acid it produces copper sulphate [X]

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$$\underset{}{CuO} + \underset{}{H_2SO_4} \rightarrow \underset{(X)}{CuSO_4} + \underset{}{H_2O}$$

in aqueous solution it exist as pentahydrate copper sulphate which is blue in colour.

$[Cu(H_2O)_5]SO_4 \cdot H_2O \rightarrow$ Blue colour

Chemical Properties of Copper Sulphate

When copper sulphate is treated with excess of aqueous solution of ammonia it produces $[Cu(NH_3)_4]^{2+}$ which has deep blue colour.

$$\underset{}{[Cu(H_2O)_4]SO_4 \cdot H_2O} + \underset{}{NH_3} \rightarrow \underset{\begin{matrix}\text{Deep blue colour}\\ \text{(neddle like crystal)}\end{matrix}}{[Cu(NH_3)_4]SO_4}$$

$[X] = [Cu(H_2O)_4]SO_4 \cdot H_2O$

$[Y] = [Cu(NH_3)_4]SO_4$

Q.4 Answer:

Their d-shells are complete

Solution:

Zn and Hg do not show variable valency like $d-$ block elements because their $d-$shell are complete their electronic confign are

$Zn^{30} \quad 3d^{10} \cdot 4s^2$

$Hg^{80} \quad 5d^{10} \cdot 6s^2$

The variable valency is shown by those elements which have got incompletely filled $'d'$ orbitals

Q.5 Answer:

The compounds formed by the $3d$ transiton metals are less ionic than the corresponding compounds formed by the 4 d and 5 d transition metals

Solution:

Due to more effective nuclear charge (because of poor shielding effect of d and f electrons) the compounds of 4 d and 5 d metals are less ionic than those of 3 d metals.

Ionization energy is defined as the amount of energy required to s the energy required to remove an electron from an isolated gaseous atom in its ground state.

Generally, ionisation enthalpy of an element increases as we go across a period and decreases as we move down the group.

Ionisation energy of transition metals

As the transition metals are present between the s-block and p-block element in the periodic table. So, the ionization energy of transition element is intermediate between s and p- block elements.

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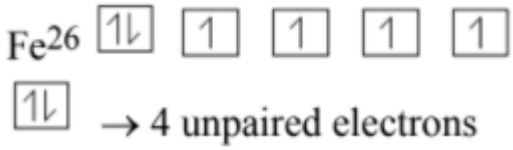
Q.6 Answer:

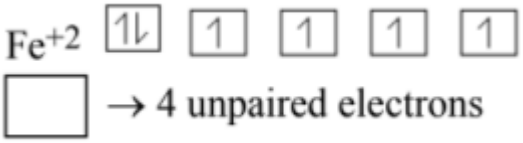
$FeSO_4, NiCl_2$

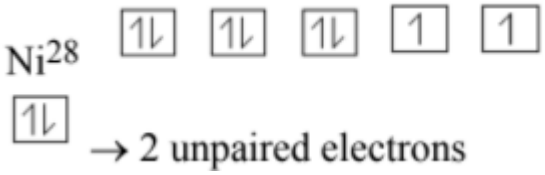
Solution:

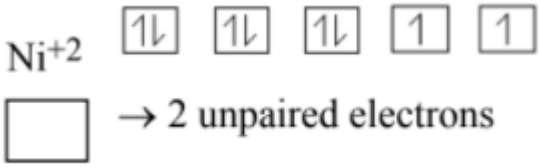
The pair which shows paramagnetism is $FeSO_4$ and $NiCl_2$. It consist of Fe^{+2} and Ni^{+2} ions which have got unpaired electrons.

Their electronic configs. are as given below.









Q.7 Answer:

reduces permanganate to Mn^{2+}

Solution:

(c) The titration of oxalic acid with $KMnO_4$ presence of HCl gives unsatisfactory result because of the fact that $KMnO_4$ can also oxidise HCl along with oxalic acid. HClon oxidation gives Cl_2 and HCl reduces $KMnO_4$ to Mn^{2+} thus the correct answer is (c).

Q.8 Answer:

Green, blue and red

Solution:

Order of strength of lighand en $> H_2O > Br$

$\therefore$ Order of Δ_0 is $\Delta_{en} > \Delta_{H_2O} > \Delta_{Br}$

$\Delta = \frac{hc}{\lambda_{absorbed}}$

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Using munsheel wheel the ralation between absorbed wavelength & colour peresived by observer is



Q.9 Answer:

None of these

Solution:

Eq. of complex = eq. of Acid = eq. of NaOH

$0.001 \times n = \frac{20 \times 0.1}{1000} \times 1$

$n = 2$

Q.10 Answer:

None of these

Solution:

(D); $\mathrm{Ni(CO)_4 - Sp^3}$ **(CO strong field ligand);**

$[\mathrm{AgF_4}]^- - \mathrm{dsp^2}\left(4\ \mathrm{d^3}\right), [\mathrm{Zn(CN)_4}]^2 - \mathrm{sp^3}\left(3\ \mathrm{d^{10}}\right)$

$[\mathrm{PtCl_4}]^{2-} - \mathrm{dsp^2}\left(5\ \mathrm{d^8}\right)$

$[\mathrm{RhCl\,(PPh_3)_3}] - \mathrm{dsp^2}\left(4\ \mathrm{d^8}\right)$

all complexes are diamagnetic in nature which show decrease in their weights in magnetic field.

Q.11 Answer:

$[\mathrm{Fe\,(H_2O)_6}]^{2+}$

Solution:

$[\mathrm{Fe\,(H_2O)_6}]^{2+}$ **has four unpaired electrons,** $[\mathrm{Cr\,(H_2O)_6}]^{3+}$ **,** $[\mathrm{Cu\,(H_2O)_6}]^{2+}$ **and** $[\mathrm{Zn\,(H_2O)_6}]^{2+}$ **have** $3, 1, 0$ **unpaired electrons respectively.**

Q.12 Answer:

$[\mathrm{Ag\,(NH_3)_2}]\,\mathrm{Cl}$

Solution:

In $[\mathrm{Ag\,(NH_3)_2}]\,\mathrm{Cl}$, $\mathrm{Ag^+}$ **contains** $\mathrm{d^{10}}$ **configuration. All others contain unpaired electrons.**

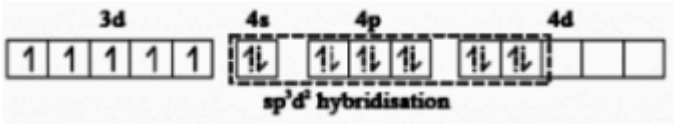
Q.13 Answer:

5.92 B.M.

Solution:

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(C); $[FeF_6]^3$ **colourles compound is formed.** F **is weak field ligand ; so,** $[FeF_6]^3$ **:**



$\mu = \sqrt{5(5+2)} = 5.92$ **B.M.**

$Fe(SCN)_3 + 3\ F^- \rightarrow FeF_3 + 3SCN^-$

Magnetic moment $= \mu = \sqrt{n + (n+2)} = \sqrt{5 \times 7} = \sqrt{35} = 5.92\text{BM}.$

Q.14 Answer:

Lanthanides are separated from one another by ion exchange methods

Solution:

According to lanthanide contraction the ionic radii of trivalent lanthanid steadily decrease with increase in atomic number.

Q.15 Answer:

The $+3$ oxidation state of cerium is more stable than the $+4$ oxidation state

Solution:

$+4$ **oxidation state of cerium is also known in solution.**

Q.16 Answer:

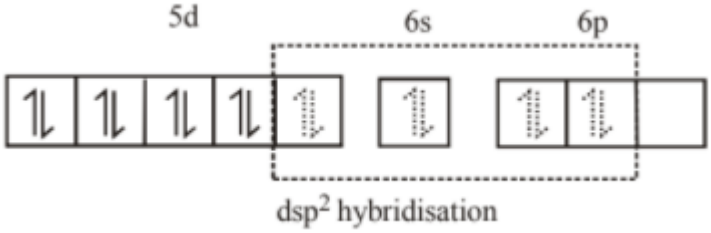
The IUPAC name of the complex is carbonlychloridobis (triphenylphosphine) iridium (I).

Solution:

(A) $[IrCl(CO)\ (Ph_3)_2]$ **: carbonylchloridobis(triphenylphosphine) iridium (I)**

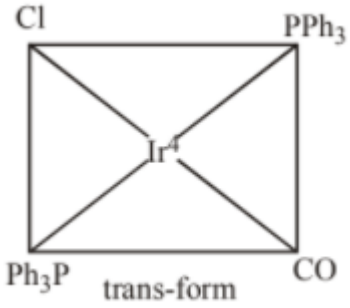
(B) Coordination number of Ir **is four.** Ir **is in** $(+1)$ **oxidation state with** $5d^8$ **configuration. So its geometry should be square planar.**

$[IrCl(CO)\ (Ph_3)_2]$

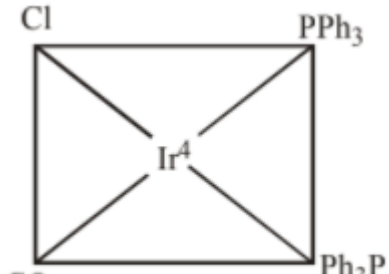


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(C) All electrons are paired, so magnetic moment is zero



trans-form



cis-form

(D) The complex has plane of symmetry, so it does not show optical isomerism.

Q.17 Answer:

220

Solution:

(a) $\Delta = h\nu = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\right)\times\left(3.0\times10^{8}\ \text{m/s}\right)}{544\times10^{-9}\ \text{m}} = 3.65\times10^{-19}\ \text{J/ion}$
$= \left(3.65\times10^{-19}\ \text{J/ion}\right)\times\left(6.02\times10^{23}\ ions/mole\right) = \frac{21.973\times10^{4}}{1000}\text{kJmol}^{-1} =$
$219.73\ \text{kJ mol}^{-1}$

(b) As it absorbs blue colour light $(\lambda = 455\ \text{nm})$**, the colour of the complex must be orange.**

(c) (i) ${}_{78}\text{Pt}$ **(II) has** 5d^8 **configuration, all electrons are paired, so diamagnetic.**

(ii) ${}_{27}\text{Co}^{2+}$ **has** 3d^7 **configuration,** SCN^- **is weak field ligand. So the complex is paramagnetic with three unpaired electrons.**

(iii) ${}_{29}\text{Cu}^{2+}$ **has** 3d^9 **configuration, complex is paramagnetic with one unpaired electron.**

(iv) ${}_{80}\text{Hg}^{2+}$ **has** 5d^{10} **configuration, all electrons are paired to diagmagnetic.**

Q.18 Answer:

A complex which has just one absorption band at $455\ \text{nm}$, must be red coloured

Solution:

(a) $\Delta = h\nu = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\right)\times\left(3.0\times10^{8}\ \text{m/s}\right)}{544\times10^{-9}\ \text{m}} = 3.65\times10^{-19}\ \text{J/ion}$
$= \left(3.65\times10^{-19}\ \text{J/ion}\right)\times\left(6.02\times10^{23}\ ions/mole\right) = \frac{21.973\times10^{4}}{1000}\text{kJmol}^{-1} =$
$219.73\ \text{kJ mol}^{-1}$

(b) As it absorbs blue colour light $(\lambda = 455\ \text{nm})$**, the colour of the complex must be orange.**

(c) (i) ${}_{78}\text{Pt}$ **(II) has** 5d^8 **configuration, all electrons are paired, so diamagnetic.**

(ii) ${}_{27}\text{Co}^{2+}$ **has** 3d^7 **configuration,** SCN^- **is weak field ligand. So the complex is paramagnetic with three**

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unpaired electrons.
(iii) $_{29}\mathrm{Cu}^{2+}$ **has** $3\mathrm{d}^{9}$ **configuration, complex is paramagnetic with one unpaired electron.**
(iv) $_{80}\mathrm{Hg}^{2+}$ **has** $5\mathrm{d}^{10}$ **configuration, all electrons are paired to diagmagnetic.**

Q.19 Answer:

(i) \& (iv)

Solution:

(a) $\Delta = \mathrm{h}v = \frac{\mathrm{hc}}{\lambda} = \frac{\left(6.626\times10^{-34}\ \mathrm{J}\right)\times\left(3.0\times10^{8}\ \mathrm{m/s}\right)}{544\times10^{-9}\ \mathrm{m}} = 3.65\times10^{-19}\ \mathrm{J/ion}$
$= \left(3.65\times10^{-19}\ \mathrm{J/ion}\right)\times\left(6.02\times10^{23}\ ions/mole\right) = \frac{21.973\times10^{4}}{1000}\mathrm{kJmol}^{-1} =$
$219.73\ \mathrm{kJ\ mol}^{-1}$
(b) As it absorbs blue colour light $(\lambda = 455\ \mathrm{nm})$**, the colour of the complex must be orange.**
(c) (i) $_{78}\mathrm{Pt}$ **(II) has** $5\mathrm{d}^{8}$ **configuration, all electrons are paired, so diamagnetic.**
(ii) $_{27}\mathrm{Co}^{2+}$ **has** $3\mathrm{d}^{7}$ **configuration,** SCN^{-} **is weak field ligand. So the complex is paramagnetic with three unpaired electrons.**
(iii) $_{29}\mathrm{Cu}^{2+}$ **has** $3\mathrm{d}^{9}$ **configuration, complex is paramagnetic with one unpaired electron.**
(iv) $_{80}\mathrm{Hg}^{2+}$ **has** $5\mathrm{d}^{10}$ **configuration, all electrons are paired to diagmagnetic.**

Q.20 Answer:

tetrathiocyanato-S-cobaltate (II) and bis (dimethylglyoximato) nickel (II)

Solution:

$(\mathrm{X}) = \left[\mathrm{Co(SCN)_4}\right]^{2-}, (\mathrm{Y}) = \left[\mathrm{Co(dmg)_2}\right]$

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